Chapter 4. Sampling Distributions

Distributions of statistics, chi-square, t and F, order statistics, multivariate normal

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4.1 Statistics and sampling distributions

A **random sample** of size n from a population with pdf f is a vector $X_1, \dots, X_n$ of i.i.d. random variables with pdf f. A **statistic** is a function $u(X_1, \dots, X_n)$ of the sample that involves no unknown parameter; its distribution is its **sampling distribution**. Most of statistical inference rests on knowing such distributions, at least approximately.

Example 4.1

For a Bernoulli(p) sample, $n\hat{p} = \sum X_i \sim B(n, p)$, so $E\hat{p} = p$ and $\operatorname{Var}\hat{p} = p(1-p)/n$. The sampling distribution of $\hat{p}$ concentrates around p as n grows, which is what makes $\hat{p}$ a sensible estimate.

Discrete case. If $\mathbf{X}$ is discrete and $Y = u(\mathbf{X})$, then

$$f_Y(y) = P(u(\mathbf{X}) = y) = \sum_{\mathbf{x}:\, u(\mathbf{x})=y} f_{\mathbf{X}}(\mathbf{x}).$$

Example 4.2

Let $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ be independent. Then

$$P(X_1 + X_2 = y) = \sum_{x=0}^{y} \frac{e^{-\lambda_1}\lambda_1^x}{x!}\frac{e^{-\lambda_2}\lambda_2^{y-x}}{(y-x)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{y!}\sum_{x=0}^{y}\binom{y}{x}\lambda_1^x\lambda_2^{y-x} = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^y}{y!},$$

so $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$. The same convolution argument shows $B(n_1, p) \oplus B(n_2, p) = B(n_1 + n_2, p)$, where $\oplus$ denotes the sum of independent variables.

4.2 Transformations of continuous variables

Motivation. Statistics are functions of the data, so we need the distribution of $Y = g(X)$. The cdf method always works; for one-to-one smooth maps the Jacobian formula gives the density directly.

Theorem 4.1 (One-to-one transformations)

Let $\mathbf{X} = (X_1, \dots, X_n)$ have pdf $f_{\mathbf{X}}$ that is positive on an open set $\mathcal{X} \subseteq \mathbb{R}^n$ and zero outside. Let $u : \mathcal{X} \to \mathcal{Y}$ be one-to-one and onto, with inverse $\mathbf{x} = w(\mathbf{y})$ continuously differentiable and Jacobian

$$J(\mathbf{y}) = \det\left(\frac{\partial w_i(\mathbf{y})}{\partial y_j}\right) \neq 0 \quad \text{on } \mathcal{Y}.$$

Then $\mathbf{Y} = u(\mathbf{X})$ has pdf $f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(w(\mathbf{y}))|J(\mathbf{y})|$ for $\mathbf{y} \in \mathcal{Y}$, and 0 otherwise.

Proof. For a set $B \subseteq \mathcal{Y}$, $\{\mathbf{Y} \in B\} = \{\mathbf{X} \in w(B)\}$, so by the change-of-variables formula for multiple integrals,

$$P(\mathbf{Y} \in B) = \int_{w(B)} f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x} = \int_B f_{\mathbf{X}}(w(\mathbf{y}))|J(\mathbf{y})|\, d\mathbf{y}.$$

Since this holds for every B, the integrand is the pdf of $\mathbf{Y}$. ∎

In one dimension, with u increasing or decreasing, this reads $f_Y(y) = f_X(w(y))|w'(y)|$.

To find the pdf of a single statistic $Y_1 = u_1(\mathbf{X})$, add auxiliary coordinates $Y_2, \dots, Y_n$ so that the full map is one-to-one, apply the theorem, and integrate the auxiliary coordinates out. The choice of auxiliary coordinates is a matter of convenience.

Theorem 4.2 (Many-to-one transformations)

Suppose $\mathcal{X}$ is, up to a set of probability zero, the disjoint union of open sets $\mathcal{X}_1, \dots, \mathcal{X}_m$, and that u restricted to each $\mathcal{X}_j$ is one-to-one from $\mathcal{X}_j$ onto $\mathcal{Y}_j$, with inverse w_j and nonvanishing Jacobian J_j. Then

$$f_{\mathbf{Y}}(\mathbf{y}) = \sum_{j:\, \mathbf{y} \in \mathcal{Y}_j} f_{\mathbf{X}}(w_j(\mathbf{y})) |J_j(\mathbf{y})|.$$

Proof. $P(\mathbf{Y} \in B) = \sum_j P(\mathbf{X} \in \mathcal{X}_j,\ u(\mathbf{X}) \in B) = \sum_j \int_{B \cap \mathcal{Y}_j} f_{\mathbf{X}}(w_j(\mathbf{y})) |J_j(\mathbf{y})|\, d\mathbf{y}$, applying the one-to-one argument of Theorem 4.1 on each piece. ∎

Example 4.3

(a) If $X \sim N(0,1)$ and $Y = X^2$, the two branches are $x = \pm\sqrt{y}$ with $|J| = 1/(2\sqrt{y})$, so

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\Gamma(1/2) 2^{1/2}} y^{1/2-1} e^{-y/2}, \qquad y > 0,$$

the Gamma$(1/2, 2)$ density.
(b) If X has pdf $f(x) = (1+x)/2$ on $(-1, 1)$ and $Y = X^2$, then for $0 < y < 1$,

$$f_Y(y) = \left\{ \frac{1+\sqrt{y}}{2} + \frac{1-\sqrt{y}}{2} \right\} \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}},$$

which is the Beta$(1/2, 1)$ density. The asymmetry of f is invisible to Y.

Location–scale families. If Z has pdf f, then $X = \mu + \sigma Z$ $(\sigma > 0)$ has pdf $\frac{1}{\sigma} f(\frac{x-\mu}{\sigma})$ by Theorem 4.1. The family $\{\frac{1}{\sigma} f(\frac{x-\mu}{\sigma}) : \mu \in \mathbb{R}, \sigma > 0\}$ is the **location–scale family** generated by f; fixing $\sigma = 1$ gives a location family and fixing $\mu = 0$ a scale family. Standard members:

family	standard pdf $f(z)$	type
normal	$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$	location–scale
logistic	$e^{-z}/(1+e^{-z})^2$	location–scale
double exponential	$\frac{1}{2} e^{-\lvert z \rvert}$	location–scale
Cauchy	$\frac{1}{\pi} \frac{1}{1+z^2}$	location–scale
exponential	e^{-z}, $z > 0$	scale
gamma (fixed α)	$z^{\alpha-1} e^{-z}/\Gamma(\alpha)$, $z > 0$	scale

In a location–scale family, $(X - \mu)/\sigma$ has a fixed distribution, so its moments and quantiles can be tabulated once. For example, $EX = \mu + \sigma EZ$ and the p-quantile is $\mu + \sigma z_p$. Similarly $U(a, b) \overset{d}{=} a + (b-a) U(0, 1)$.

Example 4.4 (Gamma to beta)

Let $X_1 \sim$ Gamma(α_1, β) and $X_2 \sim$ Gamma(α_2, β) be independent, $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1 + X_2)$. The map $(0,\infty)^2 \to (0,\infty) \times (0,1)$ has inverse $x_1 = y_1 y_2$, $x_2 = y_1(1-y_2)$,

with

$$J = \det \begin{pmatrix} y_2 & y_1 \\ 1-y_2 & -y_1 \end{pmatrix} = -y_1.$$

Hence

$$\begin{aligned} f(y_1, y_2) &= \frac{(y_1 y_2)^{\alpha_1 - 1}\{y_1(1-y_2)\}^{\alpha_2 - 1} e^{-y_1/\beta}}{\Gamma(\alpha_1)\Gamma(\alpha_2)\beta^{\alpha_1+\alpha_2}}\, y_1 \\ &= \frac{y_1^{\alpha_1+\alpha_2-1} e^{-y_1/\beta}}{\Gamma(\alpha_1+\alpha_2)\beta^{\alpha_1+\alpha_2}} \cdot \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_2^{\alpha_1-1}(1-y_2)^{\alpha_2-1}. \end{aligned}$$

The density factorizes, so $Y_1 \sim \text{Gamma}(\alpha_1+\alpha_2, \beta)$ and $Y_2 \sim \text{Beta}(\alpha_1, \alpha_2)$ are independent. Since the second factor must integrate to 1, this also proves

$$B(\alpha_1, \alpha_2) = \int_0^1 y^{\alpha_1-1}(1-y)^{\alpha_2-1} dy = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)}.$$

Example 4.5 (Dirichlet distribution)

Let $X_1, \ldots, X_{k+1}$ be independent with $X_i \sim \text{Gamma}(\alpha_i, \beta)$, $S = X_1 + \cdots + X_{k+1}$, and $Y_i = X_i/S$ for $i = 1, \ldots, k$. Use $(Y_1, \ldots, Y_k, S)$ as new coordinates: the inverse is $x_i = s y_i$ $(i \le k)$ and $x_{k+1} = s(1 - y_1 - \cdots - y_k)$. Adding the first k rows of the Jacobian matrix to the last row turns it into $(0, \ldots, 0, 1)$, so $|J| = s^k$. Therefore

$$f(y_1, \ldots, y_k, s) = \frac{\prod_{i=1}^k y_i^{\alpha_i - 1}\,(1 - \sum_{i \le k} y_i)^{\alpha_{k+1}-1}}{\prod_{i=1}^{k+1}\Gamma(\alpha_i)} \cdot \frac{s^{\alpha_1+\cdots+\alpha_{k+1}-1} e^{-s/\beta}}{\beta^{\alpha_1+\cdots+\alpha_{k+1}}}.$$

Integrating out s gives the **Dirichlet**$(\alpha_1, \ldots, \alpha_{k+1})$ density

$$f(y_1, \ldots, y_k) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_{k+1})}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_{k+1})} \prod_{i=1}^k y_i^{\alpha_i - 1} \Big(1 - \sum_{i=1}^k y_i\Big)^{\alpha_{k+1}-1}$$

on $\{y_i > 0,\ y_1 + \cdots + y_k < 1\}$, and shows that $(Y_1, \ldots, Y_k)$ is independent of S. For $k = 1$ this is $\text{Beta}(\alpha_1, \alpha_2)$.

Example 4.6 (Polar coordinates)

Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} N(0,1)$ and write $Z_1 = R\cos\Theta$, $Z_2 = R\sin\Theta$ with $R > 0$ and $0 < \Theta < 2\pi$. The Jacobian of $(r, \theta) \mapsto (z_1, z_2)$ is r, so

$$f(r, \theta) = \frac{1}{2\pi} e^{-r^2/2} r = (r e^{-r^2/2}) \cdot \frac{1}{2\pi}.$$

Thus R and Θ are independent, $\Theta \sim U(0, 2\pi)$, and $R^2/2 \sim \text{Exp}(1)$ (equivalently $R^2 \sim \chi^2(2)$). Reading this backwards, if $U_1, U_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, then $\sqrt{-2\log U_1}\cos(2\pi U_2)$ and $\sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent $N(0,1)$ variables.

Example 4.7 (A two-to-one map on the disk)

Let (X_1, X_2) be uniform on the unit disk, with pdf $1/\pi$, and put $Y_1 = X_1^2 + X_2^2$ and $Y_2 = X_1/\sqrt{Y_1}$. The map is two-to-one: on the upper and lower half disks the inverses are

$$x_1 = y_2\sqrt{y_1}, \qquad x_2 = \pm\sqrt{y_1}\sqrt{1 - y_2^2}, \qquad 0 < y_1 < 1,\ -1 < y_2 < 1.$$

A direct computation gives $|J| = 1/(2\sqrt{1-y_2^2})$ on each branch. By Theorem 4.2,

$$f(y_1, y_2) = 2 \cdot \frac{1}{\pi} \cdot \frac{1}{2\sqrt{1-y_2^2}} = 1 \cdot \frac{1}{\pi\sqrt{1-y_2^2}}.$$

So $Y_1 \sim U(0,1)$ is independent of Y_2, the cosine of the polar angle, which has the arcsine density $1/(\pi\sqrt{1-y^2})$ on $(-1,1)$.

Example 4.8

For $X_1, X_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, let $Y = (X_1+X_2)/2$ and $W = (X_1-X_2)/2$. Then $x_1 = y+w$, $x_2 = y-w$, $|J| = 2$, and (Y,W) is uniform with density 2 on $\{|w| < \min(y, 1-y)\}$. Integrating out w gives the triangular density $f_Y(y) = 4\min(y, 1-y)$ on $(0,1)$.

4.3 Sampling from a normal population

Motivation. For a normal sample, $\bar{X}$ and S^2 are independent, $\bar{X}$ is normal and $(n-1)S^2/\sigma^2$ is χ^2. These facts produce the t and F distributions and with them all exact normal-theory inference.

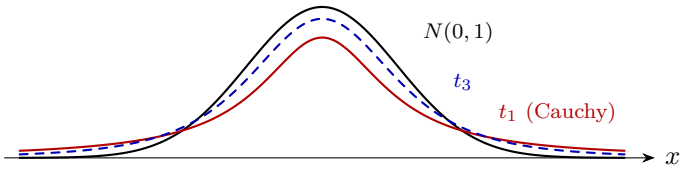


Figure. *The t densities have heavier tails than the normal; replacing σ by S costs exactly this much. As the degrees of freedom grow, $t_\nu \to N(0,1)$.*

4.3.1 The chi-square distribution

Definition (Chi-square)

For $r > 0$, $\chi^2(r) = \text{Gamma}(r/2, 2)$, with pdf

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \qquad x > 0.$$

r is the number of degrees of freedom. The upper α point $\chi^2_\alpha(r)$ is defined by $P(X > \chi^2_\alpha(r)) = \alpha$.

Theorem 4.3

(a) If $X \sim \chi^2(r)$, then $M(t) = (1-2t)^{-r/2}$ for $t < 1/2$, $EX = r$, and $\text{Var}\, X = 2r$.

(b) If $X_i \sim \chi^2(r_i)$ are independent, then $X_1 + \cdots + X_k \sim \chi^2(r_1 + \cdots + r_k)$.

(c) If $Z \sim N(0,1)$, then $Z^2 \sim \chi^2(1)$. Hence if $Z_1, \ldots, Z_r \overset{\text{i.i.d.}}{\sim} N(0,1)$, then $Z_1^2 + \cdots + Z_r^2 \sim \chi^2(r)$.

(d) $X \sim \text{Gamma}(\alpha, \beta)$ if and only if $2X/\beta \sim \chi^2(2\alpha)$.

Proof. (a) is the gamma mgf with $\alpha = r/2$, $\beta = 2$, and the moments are $\alpha\beta$ and $\alpha\beta^2$.

(b) The mgf of the sum is $\prod(1-2t)^{-r_i/2} = (1-2t)^{-\sum r_i/2}$.

(c) is Example 4.3(a) combined with (b).

(d) $2X/\beta$ has mgf $(1-\beta\cdot 2t/\beta)^{-\alpha} = (1-2t)^{-2\alpha/2}$. ■

By (d), gamma quantiles come from chi-square tables: the upper α point of Gamma(ν,β) is $\frac{\beta}{2}\chi^2_\alpha(2\nu)$.

4.3.2 The t distribution

Definition (Student's t)

If $Z\sim N(0,1)$ and $V\sim\chi^2(r)$ are independent, the distribution of $T = Z/\sqrt{V/r}$ is the t distribution with r degrees of freedom, $t(r)$. Its upper α point is $t_\alpha(r)$.

Theorem 4.4

The pdf of $t(r)$ is

$$f(t) = \frac{\Gamma(\frac{r+1}{2})}{\Gamma(\frac12)\Gamma(\frac r2)\sqrt r}\Big(1+\frac{t^2}{r}\Big)^{-\frac{r+1}{2}}, \qquad t\in\mathbb{R}.$$

It is symmetric about 0; $t(1)$ is the Cauchy distribution. $E|T|^k<\infty$ iff $k<r$, and $ET=0$ $(r>1)$, $\operatorname{Var}T = r/(r-2)$ $(r>2)$. As $r\to\infty$, $f(t)\to\phi(t)$.

Proof. Map $(z,v)\mapsto(t,v)$ with $z = t\sqrt{v/r}$; the Jacobian is $\sqrt{v/r}$. The joint pdf of (T,V) is

$$\frac{1}{\sqrt{2\pi}}e^{-t^2v/(2r)}\cdot\frac{v^{r/2-1}e^{-v/2}}{\Gamma(r/2)2^{r/2}}\cdot\sqrt{\frac vr}.$$

Integrating over $v>0$ with the gamma integral $\int_0^\infty v^{a-1}e^{-bv}dv = \Gamma(a)b^{-a}$, $a=\frac{r+1}{2}$, $b=\frac12(1+t^2/r)$, gives

$$f(t) = \frac{\Gamma(\frac{r+1}{2})2^{(r+1)/2}}{\sqrt{2\pi}\,\Gamma(\frac r2)2^{r/2}\sqrt r}\Big(1+\frac{t^2}{r}\Big)^{-\frac{r+1}{2}},$$

which is the stated form since $\Gamma(\frac12)=\sqrt\pi$. Symmetry is clear, and for $r=1$ the density is $\frac1\pi(1+t^2)^{-1}$. The tail behaves like $|t|^{-(r+1)}$, so $|t|^kf(t)$ is integrable iff $k<r$. For the variance, by independence $ET^2 = r\,EZ^2\,EV^{-1}$ and

$$EV^{-1} = \int_0^\infty \frac{v^{r/2-2}e^{-v/2}}{\Gamma(r/2)2^{r/2}}dv = \frac{\Gamma(r/2-1)2^{r/2-1}}{\Gamma(r/2)2^{r/2}} = \frac{1}{r-2}.$$

Finally $(1+t^2/r)^{-(r+1)/2}\to e^{-t^2/2}$, and the constant tends to $1/\sqrt{2\pi}$ (by Stirling's formula, or because both sides are densities and Scheffé-type arguments apply; in Chapter 5 we show $T\to N(0,1)$ in distribution directly). ■

4.3.3 Sample mean and sample variance

Theorem 4.5 (Normal samples)

Let $X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} N(\mu,\sigma^2)$, $\bar X = \frac1n\sum X_i$ and $S^2 = \frac{1}{n-1}\sum(X_i-\bar X)^2$. Then

(a) $\bar X\sim N(\mu,\sigma^2/n)$;

(b) $\bar X$ and $(X_1-\bar X,\dots,X_n-\bar X)$ are independent; in particular $\bar X$ and S^2 are independent;

(c) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$;
(d) $\sqrt{n}(\bar{X}-\mu)/S \sim t(n-1)$.

Proof. (a) and (b). Compute the joint mgf of $\bar{X}$ and the deviations. For $t, t_1, \dots, t_n$ let $\bar{t} = \frac{1}{n}\sum t_i$. Then

$$t\bar{X} + \sum_i t_i(X_i - \bar{X}) = \sum_i a_i X_i, \qquad a_i = \frac{t}{n} + (t_i - \bar{t}),$$

with $\sum a_i = t$ and $\sum a_i^2 = t^2/n + \sum (t_i - \bar{t})^2$ (the cross term vanishes because $\sum(t_i - \bar{t}) = 0$). By independence,

$$E\exp\Big(\sum a_i X_i\Big) = \prod_i \exp\Big(\mu a_i + \frac{\sigma^2 a_i^2}{2}\Big) = \exp\Big(\mu t + \frac{\sigma^2 t^2}{2n}\Big) \cdot \exp\Big(\frac{\sigma^2}{2}\sum_i (t_i - \bar{t})^2\Big).$$

The joint mgf factors into a function of t alone and a function of $(t_1, \dots, t_n)$ alone. Hence $\bar{X}$ is independent of the deviation vector, and setting $t_i = 0$ shows $\bar{X} \sim N(\mu, \sigma^2/n)$. Since S^2 is a function of the deviations, it is independent of $\bar{X}$.

(c) Expand around $\bar{X}$:

$$U := \sum_i \Big(\frac{X_i - \mu}{\sigma}\Big)^2 = \underbrace{\frac{(n-1)S^2}{\sigma^2}}_{V} + \underbrace{\Big(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\Big)^2}_{W}.$$

Here $U \sim \chi^2(n)$ and $W \sim \chi^2(1)$ by Theorem 4.3(c), and V and W are independent by (b). Therefore $M_U(t) = M_V(t) M_W(t)$, i.e.

$$M_V(t) = \frac{(1-2t)^{-n/2}}{(1-2t)^{-1/2}} = (1-2t)^{-(n-1)/2}, \qquad t < \tfrac{1}{2},$$

and $V \sim \chi^2(n-1)$ by uniqueness of mgfs.

(d) $\sqrt{n}(\bar{X}-\mu)/S = Z/\sqrt{V/(n-1)}$ with $Z = \sqrt{n}(\bar{X}-\mu)/\sigma \sim N(0,1)$ independent of $V \sim \chi^2(n-1)$. ■

Corollary 4.6 (Intervals for μ and σ^2)

For a normal sample, each of the following random intervals covers the true parameter with probability $1-\alpha$:

$$\bar{X} \pm t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \quad \text{for } \mu, \qquad \Big(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) \quad \text{for } \sigma^2.$$

Proof. By Theorem 4.5(d), $P(|\sqrt{n}(\bar{X}-\mu)/S| \le t_{\alpha/2}(n-1)) = 1-\alpha$, and the event inside is $\{\mu \in \bar{X} \pm t_{\alpha/2} S/\sqrt{n}\}$. Likewise $P(\chi^2_{1-\alpha/2} \le (n-1)S^2/\sigma^2 \le \chi^2_{\alpha/2}) = 1-\alpha$, and solving for σ^2 gives the second interval. ■

Example 4.9

A sample of $n = 16$ fill weights from a normal population gives $\bar{x} = 52.3$ and $s = 4.8$. With $t_{0.025}(15) = 2.131$, the 95% interval for μ is $52.3 \pm 2.131 \cdot 4.8/4 = 52.3 \pm 2.56$, i.e. $(49.74, 54.86)$. With $\chi^2_{0.025}(15) = 27.49$ and $\chi^2_{0.975}(15) = 6.262$, and $(n-1)s^2 = 345.6$, the interval for σ^2 is $(12.57, 55.19)$.

4.3.4 The F distribution

Definition (Snedecor's F)

If $V_1 \sim \chi^2(r_1)$ and $V_2 \sim \chi^2(r_2)$ are independent, the distribution of $F = (V_1/r_1)/(V_2/r_2)$ is $F(r_1, r_2)$. Its upper α point is $F_\alpha(r_1, r_2)$.

Theorem 4.7

(a) The pdf of $F(r_1, r_2)$ is

$$f(x) = \frac{\Gamma(\frac{r_1+r_2}{2})}{\Gamma(\frac{r_1}{2})\Gamma(\frac{r_2}{2})}\Big(\frac{r_1}{r_2}\Big)^{r_1/2} x^{r_1/2-1}\Big(1+\frac{r_1}{r_2}x\Big)^{-(r_1+r_2)/2}, \qquad x > 0.$$

(b) If $X \sim F(r_1, r_2)$, then $1/X \sim F(r_2, r_1)$; hence $F_{1-\alpha}(r_1, r_2) = 1/F_\alpha(r_2, r_1)$.

(c) If $T \sim t(r)$, then $T^2 \sim F(1, r)$; hence $t_{\alpha/2}(r)^2 = F_\alpha(1, r)$.

(d) If $X \sim F(r_1, r_2)$, then $\dfrac{(r_1/r_2)X}{1+(r_1/r_2)X} \sim \text{Beta}(r_1/2, r_2/2)$.

(e) $EX = r_2/(r_2-2)$ for $r_2 > 2$.

Proof. (d) first: $\frac{(r_1/r_2)X}{1+(r_1/r_2)X} = \frac{V_1}{V_1+V_2}$, and $V_i \sim \text{Gamma}(r_i/2, 2)$ are independent, so Example 4.4 gives $\text{Beta}(r_1/2, r_2/2)$.

(a) The map $y = cx/(1+cx)$, $c = r_1/r_2$, is increasing with $dy/dx = c/(1+cx)^2$, so

$$f(x) = \frac{1}{B(\frac{r_1}{2}, \frac{r_2}{2})}\Big(\frac{cx}{1+cx}\Big)^{r_1/2-1}\Big(\frac{1}{1+cx}\Big)^{r_2/2-1}\frac{c}{(1+cx)^2},$$

which simplifies to the stated pdf.

(b) $1/X = (V_2/r_2)/(V_1/r_1) \sim F(r_2, r_1)$, so $P(X > 1/F_\alpha(r_2, r_1)) = P(1/X < F_\alpha(r_2, r_1)) = 1-\alpha$.

(c) $T^2 = Z^2/(V/r)$ with $Z^2 \sim \chi^2(1)$ independent of V.

(e) $EX = \frac{r_2}{r_1} EV_1\, EV_2^{-1} = \frac{r_2}{r_1} \cdot r_1 \cdot \frac{1}{r_2-2}$, using EV^{-1} from the proof of Theorem 4.4. ■

Corollary 4.8 (Comparing two variances)

Let $X_{11}, \ldots, X_{1n_1} \overset{\text{i.i.d.}}{\sim} N(\mu_1, \sigma_1^2)$ and $X_{21}, \ldots, X_{2n_2} \overset{\text{i.i.d.}}{\sim} N(\mu_2, \sigma_2^2)$ be independent samples with sample variances S_1^2, S_2^2. Then

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1),$$

and a $1-\alpha$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{\alpha/2}(n_1-1, n_2-1)},\ \frac{S_1^2}{S_2^2} F_{\alpha/2}(n_2-1, n_1-1)\Big).$$

Proof. $(n_i-1)S_i^2/\sigma_i^2 \sim \chi^2(n_i-1)$ independently (Theorem 4.5), so the ratio is F by definition. Invert $P(F_{1-\alpha/2} \le \text{ratio} \le F_{\alpha/2}) = 1-\alpha$ and use Theorem 4.7(b) for the lower point. ■

4.3.5 The one-way model

Consider k independent groups,

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, k,\ j = 1, \ldots, n_i,$$

with $n = n_1 + \cdots + n_k$. Write $\bar{X}_i = \frac{1}{n_i}\sum_j X_{ij}$ for the group means, $\bar{X} = \frac{1}{n}\sum_{i,j} X_{ij} = \sum_i \frac{n_i}{n}\bar{X}_i$ for the grand mean, and $\bar{\mu} = \sum_i \frac{n_i}{n}\mu_i$.

Theorem 4.9 (One-way model)

(a) $\bar{X}_1, \ldots, \bar{X}_k$ are independent, $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$, and they are independent of

$$\mathrm{SSW} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_i)^2, \qquad \mathrm{SSW}/\sigma^2 \sim \chi^2(n-k).$$

(b) $\dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{k} n_i\{\bar{X}_i - \bar{X} - (\mu_i - \bar{\mu})\}^2 \sim \chi^2(k-1)$, independent of SSW.

(c) Consequently, with $\hat{\sigma}^2 = \mathrm{SSW}/(n-k)$, if $\mu_1 = \cdots = \mu_k$ then

$$\frac{\sum_i n_i(\bar{X}_i - \bar{X})^2/(k-1)}{\hat{\sigma}^2} \sim F(k-1, n-k).$$

Proof. (a) Apply Theorem 4.5 to each group: $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$, $\mathrm{SSW}_i/\sigma^2 := \sum_j (X_{ij} - \bar{X}_i)^2/\sigma^2 \sim \chi^2(n_i - 1)$, and $\bar{X}_i$ is independent of SSW_i. The groups are independent, so all $2k$ quantities $\bar{X}_1, \mathrm{SSW}_1, \ldots, \bar{X}_k, \mathrm{SSW}_k$ are mutually independent. Adding the chi-squares gives $\mathrm{SSW}/\sigma^2 \sim \chi^2(\sum(n_i - 1)) = \chi^2(n-k)$.

(b) Let $Y_i = \sqrt{n_i}(\bar{X}_i - \mu_i)/\sigma$, so $\mathbf{Y} = (Y_1, \ldots, Y_k)' \sim N_k(\mathbf{0}, I)$, and let $\mathbf{w} = (\sqrt{n_1/n}, \ldots, \sqrt{n_k/n})'$, a unit vector. Since $\bar{X} - \bar{\mu} = \sum_i \frac{n_i}{n}(\bar{X}_i - \mu_i)$, one checks

$$\frac{\sqrt{n_i}}{\sigma}\{\bar{X}_i - \bar{X} - (\mu_i - \bar{\mu})\} = Y_i - \sqrt{\tfrac{n_i}{n}}\,\mathbf{w}'\mathbf{Y},$$

so the quantity in (b) is $\mathbf{Y}'(I - \mathbf{w}\mathbf{w}')\mathbf{Y}$. The matrix $I - \mathbf{w}\mathbf{w}'$ is symmetric and idempotent with trace $k-1$, so the quadratic form is $\chi^2(k-1)$ by Theorem 4.18 below. It is a function of the group means only, hence independent of SSW by (a).

(c) Under equal means, $\mu_i - \bar{\mu} = 0$, and the ratio of the two independent chi-squares divided by their degrees of freedom is $F(k-1, n-k)$. ∎

For equal group sizes the argument in (b) can also be done without matrices: if $n_i \equiv m$, then $\bar{X}_i - \mu_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2/m)$ and (b) is Theorem 4.5(c) applied to these k values.

Corollary 4.10 (Two-sample t)

For $k = 2$ with common variance, let $S_p^2 = \mathrm{SSW}/(n_1 + n_2 - 2)$. Then

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t(n_1 + n_2 - 2),$$

and $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2}(n_1 + n_2 - 2) S_p\sqrt{1/n_1 + 1/n_2}$ is a $1 - \alpha$ interval for $\mu_1 - \mu_2$.

Proof. $\bar{X}_1 - \bar{X}_2 \sim N(\mu_1 - \mu_2, \sigma^2(1/n_1 + 1/n_2))$ is independent of $\mathrm{SSW}/\sigma^2 \sim \chi^2(n_1 + n_2 - 2)$ by Theorem 4.9(a); apply the definition of t. ∎

More generally, for any contrast $\sum c_i\mu_i$, $\left(\sum c_i\bar{X}_i - \sum c_i\mu_i\right)/\left(\hat{\sigma}\sqrt{\sum c_i^2/n_i}\right) \sim t(n-k)$.

4.4 Order statistics

Let $X_1, \dots, X_n$ be a random sample from a continuous distribution with pdf f and cdf F. Ties occur with probability 0, so the sorted values $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$, the **order statistics**, are well defined. The sample median, range $X_{(n)} - X_{(1)}$, and extremes are functions of them.

Theorem 4.11

The joint pdf of $(X_{(1)}, \dots, X_{(n)})$ is

$$f_{(1,\dots,n)}(y_1, \dots, y_n) = n!\, f(y_1) f(y_2) \cdots f(y_n), \qquad y_1 < y_2 < \cdots < y_n.$$

Proof. Up to a null set, $\mathbb{R}^n$ is the disjoint union of the $n!$ open sets $\mathcal{X}_\pi = \{x_{\pi(1)} < \cdots < x_{\pi(n)}\}$, π a permutation. On each $\mathcal{X}_\pi$ the sorting map $\mathbf{x} \mapsto (x_{(1)}, \dots, x_{(n)})$ is the coordinate permutation $\mathbf{x} \mapsto (x_{\pi(1)}, \dots, x_{\pi(n)})$, one-to-one onto $\{y_1 < \cdots < y_n\}$ with $|J| = 1$. The joint pdf of the sample is symmetric, $\prod f(x_i)$, so each branch contributes $\prod f(y_i)$, and Theorem 4.2 gives $n! \prod f(y_i)$. ∎

Theorem 4.12

(a) The cdf of $X_{(r)}$ is $P(X_{(r)} \le y) = \sum_{j=r}^{n} \binom{n}{j} F(y)^j \{1 - F(y)\}^{n-j}$, and its pdf is

$$f_{(r)}(y) = \frac{n!}{(r-1)!\,(n-r)!} F(y)^{r-1} \{1 - F(y)\}^{n-r} f(y).$$

(b) For $r < s$, the joint pdf of $(X_{(r)}, X_{(s)})$ is, for $y < z$,

$$f_{(r,s)}(y, z) = \frac{n!}{(r-1)!(s-r-1)!(n-s)!} \times F(y)^{r-1} \{F(z) - F(y)\}^{s-r-1} \{1 - F(z)\}^{n-s} f(y) f(z).$$

In particular $X_{(n)}$ has cdf F^n, and $X_{(1)}$ has cdf $1 - (1 - F)^n$.

Proof. (a) $X_{(r)} \le y$ iff at least r observations are $\le y$, and the number $N(y)$ of observations $\le y$ is $B(n, F(y))$. This gives the cdf. Differentiate it, writing $p = F(y)$:

$$\frac{d}{dp} \sum_{j=r}^{n} \binom{n}{j} p^j (1-p)^{n-j} = \sum_{j=r}^{n} \left\{ n\tbinom{n-1}{j-1} p^{j-1} (1-p)^{n-j} - n\tbinom{n-1}{j} p^j (1-p)^{n-1-j} \right\}.$$

The sum telescopes: the negative part of the j-th term cancels the positive part of the $(j+1)$-th term, and the last negative term is zero. Only $n\binom{n-1}{r-1} p^{r-1} (1-p)^{n-r}$ remains. Multiply by $f(y)$ (chain rule).

(b) Integrate the joint pdf of Theorem 4.11 over the other coordinates. For $y_r = y$ and $y_s = z$ fixed, the variables below y contribute

$$\int_{y_1 < \cdots < y_{r-1} < y} \prod_{i<r} f(y_i)\, dy_1 \cdots dy_{r-1} = \frac{F(y)^{r-1}}{(r-1)!},$$

since by symmetry the unordered integral $F(y)^{r-1}$ is split equally among the $(r-1)!$ orderings. Similarly the variables between y and z give $\{F(z) - F(y)\}^{s-r-1}/(s-r-1)!$, and those above z give $\{1 - F(z)\}^{n-s}/(n-s)!$. Multiplying by $n! f(y) f(z)$ gives (b). The same computation with one fixed coordinate reproves (a). ∎

The constants have a multinomial reading: for $X_{(r)} \in (y, y + dy]$ we need $r - 1$ observations below y, one in $(y, y + dy]$ and $n - r$ above y.

Example 4.10 (Uniform order statistics)

For a $U(0,1)$ sample, $F(y) = y$, so $X_{(r)} \sim \text{Beta}(r, n-r+1)$, with

$$EX_{(r)} = \frac{r}{n+1}, \qquad \text{Var}\, X_{(r)} = \frac{r(n-r+1)}{(n+1)^2(n+2)}.$$

For instance, with $n = 5$, $P(X_{(3)} > 0.7)$ is the probability that at most two observations fall below 0.7:

$$\sum_{j=0}^{2} \binom{5}{j} (0.7)^j (0.3)^{5-j} = 0.0024 + 0.0284 + 0.1323 = 0.163.$$

The joint pdf of $(X_{(1)}, X_{(n)})$ is $n(n-1)(z-y)^{n-2}$ on $0 < y < z < 1$. Changing to $R = X_{(n)} - X_{(1)}$, $M = X_{(1)}$ ($|J| = 1$) and integrating m over $(0, 1-r)$ gives the range density $n(n-1)r^{n-2}(1-r)$, i.e. $R \sim \text{Beta}(n-1, 2)$.

Example 4.11 (Uniform spacings)

Let $U_{(1)} < \cdots < U_{(n)}$ be uniform order statistics and $D_1 = U_{(1)}$, $D_i = U_{(i)} - U_{(i-1)}$ $(2 \le i \le n)$. The map is linear and triangular with unit diagonal, so $|J| = 1$ and $(D_1, \ldots, D_n)$ has density $n!$ on $\{d_i > 0,\ d_1 + \cdots + d_n < 1\}$. This is the Dirichlet$(1, \ldots, 1)$ density with $n+1$ parameters, since $\Gamma(n+1)/\Gamma(1)^{n+1} = n!$. By Example 4.5, $(D_1, \ldots, D_n) \overset{d}{=} (E_1, \ldots, E_n)/(E_1 + \cdots + E_{n+1})$ for $E_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$. In particular, every spacing (including $1 - U_{(n)}$) has the same Beta$(1, n)$ distribution.

Example 4.12 (Exponential spacings)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\beta)$, with order statistics $Y_i = X_{(i)}$ and $Y_0 = 0$. Since $\sum_i y_i = \sum_{i=1}^n (n-i+1)(y_i - y_{i-1})$, the joint pdf $n!\beta^{-n} e^{-\sum y_i/\beta}$ becomes, in the variables $W_i = (n-i+1)(Y_i - Y_{i-1})$ (Jacobian of the inverse $1/n!$),

$$f(w_1, \ldots, w_n) = \prod_{i=1}^{n} \frac{1}{\beta} e^{-w_i/\beta}, \qquad w_i > 0.$$

So the normalized spacings $W_1, \ldots, W_n$ are i.i.d. Exp(β). Consequently $X_{(1)} \sim \text{Exp}(\beta/n)$, and

$$X_{(r)} = \sum_{i=1}^{r} \frac{W_i}{n-i+1}, \qquad EX_{(r)} = \beta \sum_{i=1}^{r} \frac{1}{n-i+1}, \qquad \text{Var}\, X_{(r)} = \beta^2 \sum_{i=1}^{r} \frac{1}{(n-i+1)^2}.$$

This is the memoryless property in action: after the $(i-1)$-th failure, the remaining $n-i+1$ lifetimes start afresh.

Theorem 4.13 (Probability integral transform)

Let F be a cdf and $F^{-1}(u) = \inf\{x : F(x) \ge u\}$ for $0 < u < 1$.

(a) If $U \sim U(0,1)$, then $F^{-1}(U) \sim F$.

(b) If $X \sim F$ and F is continuous, then $F(X) \sim U(0,1)$.

Proof. (a) By right continuity, $F(F^{-1}(u)) \ge u$, and $F^{-1}(u) \le x$ iff $u \le F(x)$. Hence $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$.

(b) For $0 < u < 1$, let $x_u = F^{-1}(u)$. By continuity $F(x_u) = u$, and $F(X) \geq u$ iff $X \geq x_u$ except on a set of probability 0 (where F is flat). So $P(F(X) \geq u) = P(X \geq x_u) = 1 - F(x_u) = 1 - u$, i.e. $P(F(X) < u) = u$ for all $u \in (0, 1)$, which identifies $U(0, 1)$. ■

Corollary 4.14

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of a $U(0, 1)$ sample.

(a) For any cdf F, $(X_{(1)}, \ldots, X_{(n)}) \stackrel{d}{=} (F^{-1}(U_{(1)}), \ldots, F^{-1}(U_{(n)}))$.

(b) If F is continuous, then $(F(X_{(1)}), \ldots, F(X_{(n)})) \stackrel{d}{=} (U_{(1)}, \ldots, U_{(n)})$; in particular $F(X_{(r)}) \sim \text{Beta}(r, n - r + 1)$ whatever F is.

Proof. F^{-1} and F are nondecreasing, so they carry the sorted sample to the sorted transformed sample. Apply Theorem 4.13 coordinatewise to the i.i.d. sample and sort. ■

Example 4.13 (Distribution-free interval for a median)

Let m be the median of a continuous F. Then $X_{(r)} < m$ iff at least r observations are below m, and each observation is below m with probability $1/2$. So for $r < s$,

$$P(X_{(r)} < m < X_{(s)}) = \sum_{j=r}^{s-1} \binom{n}{j} 2^{-n},$$

which does not depend on F. With $n = 10$, $r = 2$, $s = 9$, this is $(1024-1-10-10-1)/1024 = 0.979$.

4.5 The multivariate normal distribution

Throughout, vectors are columns, $\mathbf{Z} = (Z_1, \ldots, Z_m)'$ has i.i.d. $N(0, 1)$ entries, written $\mathbf{Z} \sim N_m(\mathbf{0}, I)$, and Σ denotes a symmetric nonnegative definite matrix.

Definition (Multivariate normal)

A random vector $\mathbf{X} \in \mathbb{R}^k$ has the **multivariate normal** distribution $N_k(\mu, \Sigma)$ if any of the following equivalent conditions holds:

(1) $\mathbf{X} \stackrel{d}{=} A\mathbf{Z} + \mu$ for some $k \times m$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_m(\mathbf{0}, I)$;

(2) the mgf of $\mathbf{X}$ is $M(\mathbf{t}) = \exp\left(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}\right)$ for all $\mathbf{t} \in \mathbb{R}^k$;

(3) for every $\mathbf{t}$, $\mathbf{t}'\mathbf{X} \sim N(\mathbf{t}'\mu, \mathbf{t}'\Sigma\mathbf{t})$;

(4) (when Σ is nonsingular) $\mathbf{X}$ has pdf

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{k/2}(\det\Sigma)^{1/2}} \exp\Big\{-\frac{1}{2}(\mathbf{x} - \mu)'\Sigma^{-1}(\mathbf{x} - \mu)\Big\}.$$

Theorem 4.15

The four conditions are equivalent (the fourth under $\det\Sigma > 0$). Moreover, if $\mathbf{X} \sim N_k(\mu, \Sigma)$, then $E\mathbf{X} = \mu$ and $\operatorname{Var}\mathbf{X} = \Sigma$.

Proof. (1)⇒(2): $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}Ee^{(A'\mathbf{t})'\mathbf{Z}} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, using the mgf $e^{\mathbf{s}'\mathbf{s}/2}$ of $\mathbf{Z}$.

(2)⇒(3): the mgf of $\mathbf{t}'\mathbf{X}$ at s is $M(s\mathbf{t}) = \exp(s\mathbf{t}'\mu + \frac{1}{2}s^2\mathbf{t}'\Sigma\mathbf{t})$, the $N(\mathbf{t}'\mu, \mathbf{t}'\Sigma\mathbf{t})$ mgf.

(3)⇒(2): $M(\mathbf{t}) = Ee^{\mathbf{t}'\mathbf{X}}$ is the mgf of the normal variable $\mathbf{t}'\mathbf{X}$ evaluated at 1, i.e. $\exp(\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$; here μ and Σ are the mean and covariance of $\mathbf{X}$, because $E\mathbf{t}'\mathbf{X} = \mathbf{t}'E\mathbf{X}$ and $\operatorname{Var}\mathbf{t}'\mathbf{X} = \mathbf{t}'\operatorname{Var}(\mathbf{X})\mathbf{t}$ for all $\mathbf{t}$.

(2)⇒(1): take $A = \Sigma^{1/2}$ (or any A with $AA' = \Sigma$); then $A\mathbf{Z} + \mu$ has the same mgf, and mgfs determine distributions.

(1)⇔(4): if Σ is nonsingular, take $A = \Sigma^{1/2}$, which is invertible. The map $\mathbf{z} \mapsto \mathbf{x} = A\mathbf{z} + \mu$ has inverse Jacobian $\det A^{-1} = (\det\Sigma)^{-1/2}$, and $\mathbf{z}'\mathbf{z} = (\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$. Theorem 4.1 applied to $\prod\phi(z_i) = (2\pi)^{-k/2}e^{-\mathbf{z}'\mathbf{z}/2}$ gives the pdf in (4). Conversely a vector with pdf (4) has the same distribution as $\Sigma^{1/2}\mathbf{Z} + \mu$.

Finally, the cgf $\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}$ has gradient μ and Hessian Σ at $\mathbf{t} = \mathbf{0}$. ■

Example 4.14 (Bivariate normal)

For $k = 2$, write $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ with $|\rho| < 1$. Then $\det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2)$, and with $z_i = (x_i - \mu_i)/\sigma_i$,

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\Big\{ -\frac{z_1^2 - 2\rho z_1 z_2 + z_2^2}{2(1-\rho^2)} \Big\}.$$

Completing the square, $z_1^2 - 2\rho z_1 z_2 + z_2^2 = (1-\rho^2)z_1^2 + (z_2 - \rho z_1)^2$, so f is the $N(\mu_1, \sigma_1^2)$ density of x_1 times the $N\big(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1),\ \sigma_2^2(1-\rho^2)\big)$ density of x_2.

Theorem 4.16 (Properties)

Let $\mathbf{X} \sim N_k(\mu, \Sigma)$.

(a) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_k(\mathbf{0}, I)$.

(b) For a $p \times k$ matrix B and $\mathbf{b} \in \mathbb{R}^p$, $B\mathbf{X} + \mathbf{b} \sim N_p(B\mu + \mathbf{b}, B\Sigma B')$.

(c) If $\mathbf{X}_1 \sim N_k(\mu_1, \Sigma_1)$ and $\mathbf{X}_2 \sim N_k(\mu_2, \Sigma_2)$ are independent, then $\mathbf{X}_1 + \mathbf{X}_2 \sim N_k(\mu_1 + \mu_2, \Sigma_1 + \Sigma_2)$.

(d) For matrices A $(p \times k)$ and B $(q \times k)$, $A\mathbf{X}$ and $B\mathbf{X}$ are independent iff $\operatorname{Cov}(A\mathbf{X}, B\mathbf{X}) = A\Sigma B' = 0$.

Proof. (a) is (2)⇒(1) above.

(b) $B\mathbf{X} + \mathbf{b} \overset{d}{=} (BA)\mathbf{Z} + (B\mu + \mathbf{b})$ with $(BA)(BA)' = B\Sigma B'$.

(c) Multiply the mgfs.

(d) Independence always implies zero covariance. Conversely, the stacked vector $\binom{A\mathbf{X}}{B\mathbf{X}}$ is normal by (b) with covariance $\begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix}$. If $A\Sigma B' = 0$, then its mgf at $(\mathbf{s}, \mathbf{t})$ is

$$\exp\left(\mathbf{s}'A\mu + \tfrac{1}{2}\mathbf{s}'A\Sigma A'\mathbf{s}\right) \cdot \exp\left(\mathbf{t}'B\mu + \tfrac{1}{2}\mathbf{t}'B\Sigma B'\mathbf{t}\right),$$

the product of the two marginal mgfs. So $A\mathbf{X}$ and $B\mathbf{X}$ are independent. ■

Taking A and B to be coordinate selections: subvectors of a multivariate normal vector are independent iff their cross-covariance block is zero. This needs joint normality. If $Z \sim N(0,1)$ and

$W = SZ$ with an independent random sign $S = \pm 1$, then $W \sim N(0,1)$ and $\operatorname{Cov}(Z,W) = 0$, but $|W| = |Z|$, so Z and W are dependent; (Z,W) is not bivariate normal, since $P(Z+W=0) = 1/2$.

Theorem 4.17 (Marginal and conditional distributions)

Partition

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right),$$

with Σ_{22} nonsingular. Then

(a) $\mathbf{X}_1 \sim N(\mu_1, \Sigma_{11})$ and $\mathbf{X}_2 \sim N(\mu_2, \Sigma_{22})$;

(b) $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2),\ \Sigma_{11\cdot 2})$, where $\Sigma_{11\cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

Proof. (a) Apply Theorem 4.16(b) with $B = (I\ 0)$ or $(0\ I)$.

(b) Use the block transformation

$$\begin{pmatrix} \mathbf{W} \\ \mathbf{X}_2 \end{pmatrix} = \begin{pmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}.$$

By Theorem 4.16(b), $(\mathbf{W}, \mathbf{X}_2)$ is normal with

$$\operatorname{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0, \qquad \operatorname{Var}\mathbf{W} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

So $\mathbf{W} \sim N(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11\cdot 2})$ is independent of $\mathbf{X}_2$ by part (d) of that theorem. Since $\mathbf{X}_1 = \mathbf{W} + \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ only shifts the (unchanged) law of $\mathbf{W}$ by $\Sigma_{12}\Sigma_{22}^{-1}\mathbf{x}_2$. ■

Two features stand out: the conditional mean is linear in $\mathbf{x}_2$ (the regression of $\mathbf{X}_1$ on $\mathbf{X}_2$ is linear), and the conditional covariance does not depend on $\mathbf{x}_2$.

Example 4.15

Let $(X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

For the law of X_1 given $(X_2, X_3) = (x_2, x_3)$: $\Sigma_{12} = (2,1)$ and $\Sigma_{22}^{-1} = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, so $\Sigma_{12}\Sigma_{22}^{-1} = \frac{1}{5}(3,1)$. The conditional mean is $(3x_2 + x_3)/5$ and the conditional variance is $4 - \frac{1}{5}(3\cdot 2 + 1\cdot 1) = 13/5$. Also $X_1 - X_2$ and $X_1 + X_2 - 3X_3$ are independent iff $(1,-1,0)\Sigma(1,1,-3)' = 0$; here $\Sigma(1,1,-3)' = (3,2,-4)'$, and the inner product is $3 - 2 = 1 \neq 0$, so they are dependent.

4.5.1 Quadratic forms

Theorem 4.18

(a) If $\mathbf{X} \sim N_k(\mu, \Sigma)$ with Σ nonsingular, then $(\mathbf{X} - \mu)'\Sigma^{-1}(\mathbf{X} - \mu) \sim \chi^2(k)$.

(b) Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and A symmetric. Then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ if and only if $A^2 = A$, and in that case $r = \operatorname{rank} A = \operatorname{tr} A$.

(c) If A and B are symmetric idempotent with $AB = 0$, then $\mathbf{Z}'A\mathbf{Z}$ and $\mathbf{Z}'B\mathbf{Z}$ are independent.

Proof. (a) $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \mu) \sim N_k(\mathbf{0}, I)$ and the form equals $\mathbf{Z}'\mathbf{Z} = \sum Z_i^2$.

(b) Diagonalize $A = P'\Lambda P$ with P orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then $\mathbf{Y} = P\mathbf{Z} \sim N_n(\mathbf{0}, PP') = N_n(\mathbf{0}, I)$ and $\mathbf{Z}'A\mathbf{Z} = \sum \lambda_i Y_i^2$, whose mgf is $\prod_i (1 - 2\lambda_i t)^{-1/2}$ near $t = 0$. If $A^2 = A$, every $\lambda_i \in \{0, 1\}$, the number of ones is $\operatorname{tr} A = \operatorname{rank} A = r$, and the form is a sum of r squared independent standard normals. Conversely, if the form is $\chi^2(r)$, then $\prod_i (1 - 2\lambda_i t) = (1 - 2t)^r$ for all t near 0. Both sides are polynomials in t, so they agree everywhere and have the same roots: every nonzero λ_i equals 1, and there are r of them. Then $\Lambda^2 = \Lambda$ and $A^2 = A$.

(c) $A\mathbf{Z}$ and $B\mathbf{Z}$ are jointly normal with $\operatorname{Cov}(A\mathbf{Z}, B\mathbf{Z}) = AB' = AB = 0$, hence independent by Theorem 4.16(d). Since A is symmetric idempotent, $\mathbf{Z}'A\mathbf{Z} = (A\mathbf{Z})'(A\mathbf{Z})$ is a function of $A\mathbf{Z}$; likewise for B. ∎

The ellipsoid $\{\mu : (\mathbf{X} - \mu)'\Sigma^{-1}(\mathbf{X} - \mu) \le \chi^2_\alpha(k)\}$ is therefore a $1 - \alpha$ confidence region for μ when Σ is known.

Example 4.16 (Sample variance revisited)

Let $\mathbf{1}$ be the vector of ones and $J_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, the projection onto the span of $\mathbf{1}$. For $\mathbf{X} \sim N_n(\mu\mathbf{1}, \sigma^2 I)$, put $\mathbf{Z} = (\mathbf{X} - \mu\mathbf{1})/\sigma$. Then

$$\frac{(n-1)S^2}{\sigma^2} = \mathbf{Z}'(I - J_n)\mathbf{Z}, \qquad \frac{n(\bar{X} - \mu)^2}{\sigma^2} = \mathbf{Z}' J_n \mathbf{Z},$$

because $(I - J_n)\mathbf{1} = \mathbf{0}$. Both matrices are symmetric idempotent, $(I - J_n)J_n = 0$, and $\operatorname{tr}(I - J_n) = n - 1$. Theorem 4.18 gives parts (b) and (c) of Theorem 4.5 again.

4.5.2 The normal linear model

Let $\mathbf{Y} = X\beta + \mathbf{e}$, where X is a known $n \times p$ matrix of rank p, $\beta \in \mathbb{R}^p$ is unknown, and $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$. The least squares estimator and the residual variance estimator are

$$\hat{\beta} = (X'X)^{-1}X'\mathbf{Y}, \qquad \hat{\sigma}^2 = \frac{1}{n - p}\|\mathbf{Y} - X\hat{\beta}\|^2.$$

Theorem 4.19

(a) $\hat{\beta} \sim N_p(\beta, \sigma^2(X'X)^{-1})$.

(b) $(n - p)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n - p)$, independent of $\hat{\beta}$.

(c) For each j, with c_{jj} the j-th diagonal entry of $(X'X)^{-1}$, $(\hat{\beta}_j - \beta_j)/(\hat{\sigma}\sqrt{c_{jj}}) \sim t(n - p)$.

Proof. Let $H = X(X'X)^{-1}X'$, the projection onto the column space of X; it is symmetric, idempotent, of trace $\operatorname{tr}\{(X'X)^{-1}X'X\} = p$, and $HX = X$.

(a) $\hat{\beta} = (X'X)^{-1}X'\mathbf{Y}$ is linear in $\mathbf{Y} \sim N(X\beta, \sigma^2 I)$, with mean $(X'X)^{-1}X'X\beta = \beta$ and covariance $\sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) The residual vector is $\mathbf{Y} - X\hat{\beta} = (I - H)\mathbf{Y} = (I - H)\mathbf{e}$, since $(I - H)X = 0$. Hence $(n-p)\hat{\sigma}^2/\sigma^2 = \mathbf{Z}'(I - H)\mathbf{Z}$ with $\mathbf{Z} = \mathbf{e}/\sigma \sim N_n(\mathbf{0}, I)$, and $I - H$ is idempotent with trace $n - p$; Theorem 4.18(b) applies. For independence, $\operatorname{Cov}\left((X'X)^{-1}X'\mathbf{Y}, (I - H)\mathbf{Y}\right) = \sigma^2(X'X)^{-1}X'(I - H) = 0$, so $\hat{\beta}$ is independent of the residual vector (Theorem 4.16(d)), hence of $\hat{\sigma}^2$.

(c) Combine (a) and (b) in the definition of t. ∎

Example 4.17 (Simple linear regression)

For $Y_i = \beta_0 + \beta_1 x_i + e_i$, the estimators are

$$\hat{\beta}_1 = \frac{S_{xY}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}, \qquad \hat{\beta}_1 \sim N\Big(\beta_1, \frac{\sigma^2}{S_{xx}}\Big),$$

where $S_{xx} = \sum (x_i - \bar{x})^2$ and $S_{xY} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$. The residual estimator is $\hat{\sigma}^2 = (S_{YY} - S_{xY}^2/S_{xx})/(n-2)$, with $S_{YY} = \sum (Y_i - \bar{Y})^2$. By Theorem 4.19,

$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t(n-2).$$

With $n = 10$, $S_{xx} = 40$, $\hat{\beta}_1 = 1.5$ and $\hat{\sigma}^2 = 2.0$, the standard error is $\sqrt{2/40} = 0.224$, and with $t_{0.025}(8) = 2.306$ the 95% interval for the slope is 1.5 ± 0.52.

Summary of sampling distributions

distribution	representation
$\chi^2(r) = \text{Gamma}(r/2, 2)$	$Z_1^2 + \cdots + Z_r^2$, $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$
$t(r)$	$Z/\sqrt{V/r}$, $Z \sim N(0,1) \perp V \sim \chi^2(r)$
$F(r_1, r_2)$	$(V_1/r_1)/(V_2/r_2)$, $V_i \sim \chi^2(r_i)$ independent
$\text{Beta}(\alpha_1, \alpha_2)$	$Z_1/(Z_1+Z_2)$, $Z_i \sim \text{Gamma}(\alpha_i, \beta)$ independent
$\text{Dirichlet}(\alpha_1, \ldots, \alpha_{k+1})$	$(Z_1, \ldots, Z_k)/(Z_1 + \cdots + Z_{k+1})$, $Z_i \sim \text{Gamma}(\alpha_i, \beta)$
$N_k(\mu, \Sigma)$	$\Sigma^{1/2}\mathbf{Z} + \mu$, $\mathbf{Z} \sim N_k(\mathbf{0}, I)$